

Setup and Hold time in D-Flip Flops: Finding the ideal point to deliver time and input signal avoiding metastability and time waste

Ricardo Landeros Aranda | ECEN 4392 Numerical Methods in Python for Engineers | Spring 2026

Instructor: Marcus Mellor

Description: I propose to build a Python tool that numerically finds the setup and hold time constraints of a flip-flop using SPICE simulation data and 1D/2D root-finding methods, with an emphasis on bisection method and/or other numerical methods. The current characterization approach sweeps a 2D grid of setup and hold values and tests each point by running a SPICE simulation to measure the clock-to-Q delay.

The proposed project will reformulate the problem as a numerical method for a root-finding task: For 1D problems I will approach with the bisection method. For 2D problems I will utilize a combination of bisection along the lines and numerical contour extraction.

Goals:

1. Model the flip flop timing constraint as a numerical method root-finding problem.
2. Implement bisection-based search for 1D constraints.
3. Extend the method to approximate a 2D setup-hold contour.
4. Compare numerical efficiency and accuracy vs. linear sweep.

Requirements:

1. Correct bisection for 1D toy functions.
2. 1D timing constraints solver with tolerance.
3. 2D contour approximation quality.
4. Evaluation count with improvement over linear sweep.
5. Automated tests for numerical correctness.